

W M+ Data / Reporting / Intelligence Service Layer

v1 Scope - FL-independent portion

STATUS: DRAFT FOR APPROVAL - no service code until approved

2026-09-28

1. Standing design policy (adopted)

- Authoritative server services are built once. Pages, UFS, reports, Watch, Ask M+, voice and automation consume the same calculations.
- No separate pieces of code independently calculating the same accounting answer.
- Ask M+ is one consumer of the layer, not the layer itself.
- Reports follow the same policy: a report consumes the same authoritative service calculation an operational screen or Ask M+ would consume. Escrow/cash-flow reuse is the model, not the exception.

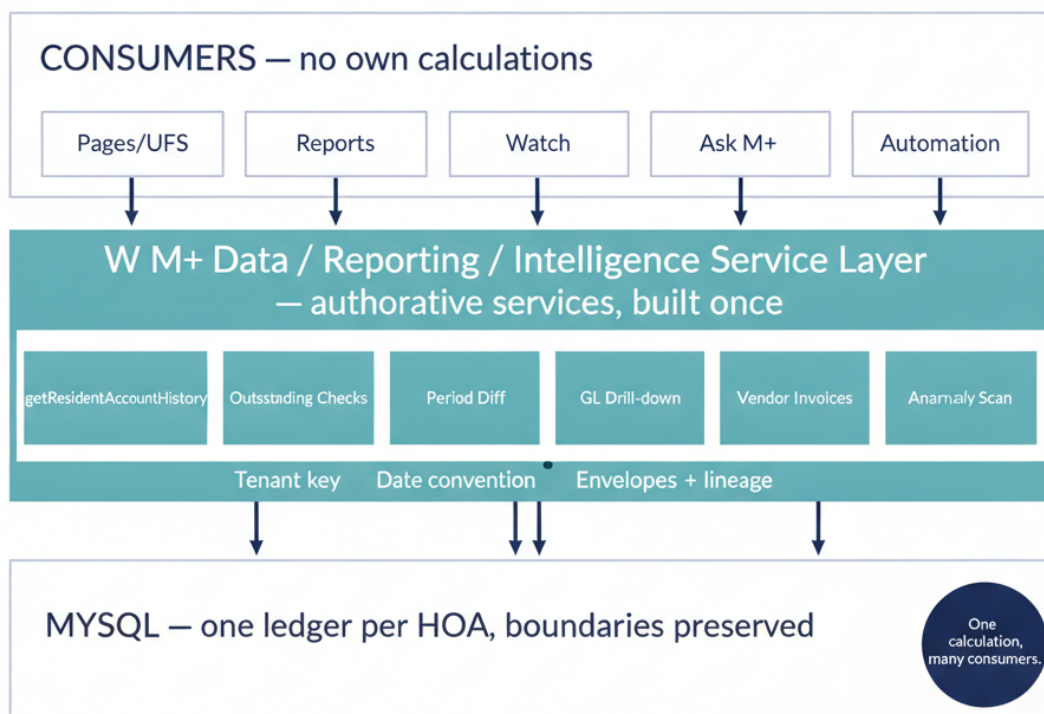
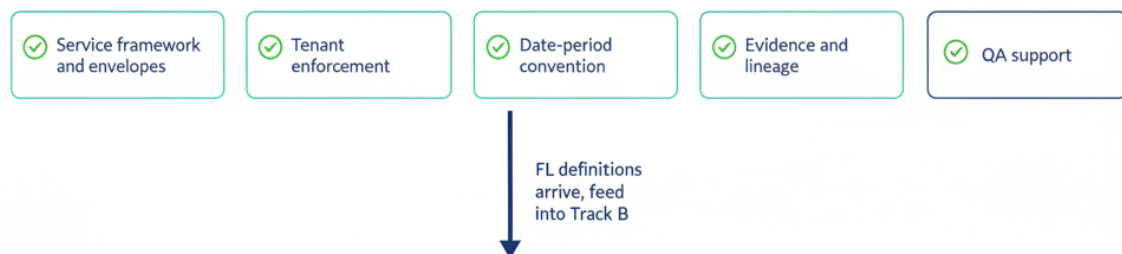


Fig. 1 - One calculation, many consumers.

2. Two tracks

- TRACK A (proceed now): service framework, tenant enforcement, date/range filters, evidence/lineage, event/status access, result/summary envelopes, QA support.
- TRACK B (waits for FL definitions): outstanding-violations/actions, getResidentFLHistory(), Manage Violations / Late Assessments / Arrears.

TRACK A – proceed now (FL-independent)



TRACK B – waits for FL definitions



Fig. 2 - Track A proceeds; Track B waits for FL definitions.

3. v1 function list - the eight questions (locked)

1. What does this resident owe? [A] getResidentAccountHistory (balance)
2. What payments produced this balance? [A] getResidentAccountHistory (provenance)
3. Which checks remain outstanding? [A] getOutstandingChecks
4. What changed between two periods? [A] getPeriodDiff (needs date convention)
5. Which transactions belong to this GL#? [A] getGLTransactions
6. What violations/actions are outstanding? [B] Blocked - needs FL definitions
7. What invoices belong to this vendor? [A] getVendorInvoices
8. What is unusual or overdue? [A*] Anomaly framework; overdue follows FL later

Q1+Q2 are one function, not two: balance + the payments that produced it + evidence references. Q8 carries a boundary: the generic anomaly framework proceeds in Track A, while assessment-overdue thresholds follow FL rules when defined.

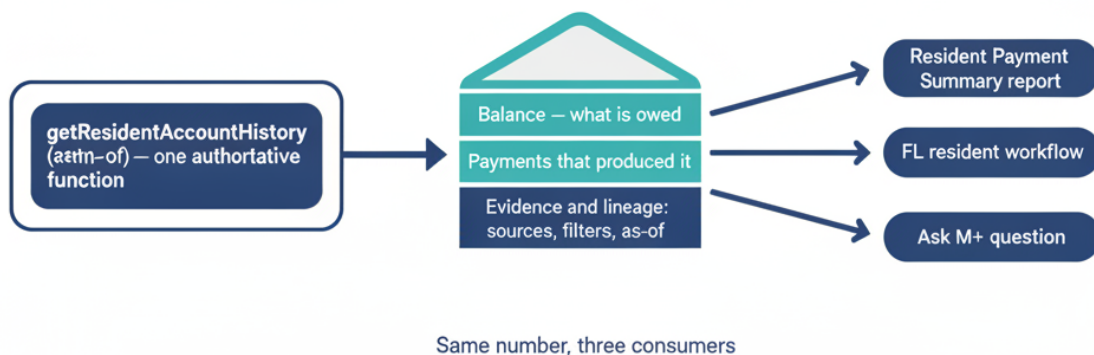


Fig. 3 - getResidentAccountHistory: same number, three consumers.

4. Tenant key (adopted)

Authoritative tenant/HOA key: (MgtCoClientID, HOALicenseNumber) - not the ClientID shorthand. Cross-HOA aggregation, when authorized, preserves individual HOA/ledger boundaries.

5. Common conventions (need sign-off)

- Date/period filtering: one convention for as-of dates, ranges and fiscal periods.
- Standard envelopes: { result rows, summary, lineage/evidence refs, filters applied } + errors + pagination.
- Evidence/lineage: every answer carries its sources so all consumers cite the same basis.

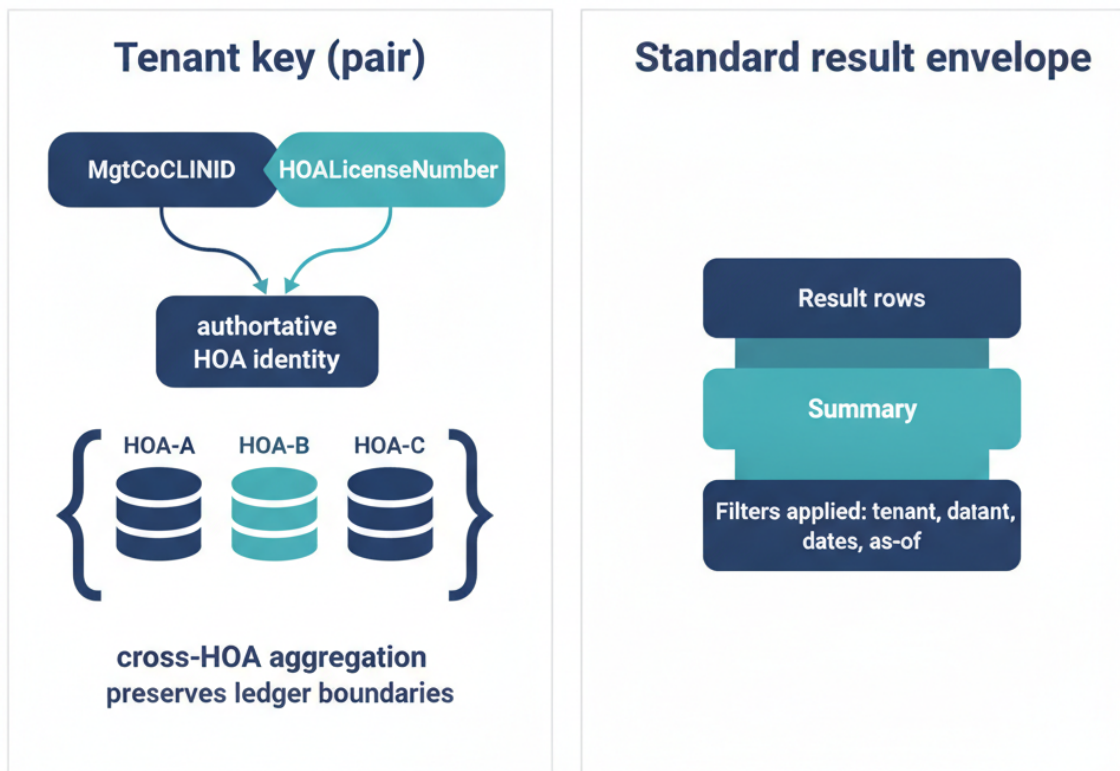


Fig. 4 - Tenant pair and standard result envelope.

6. Add while building (adopted)

Append-only FL event history, sent-letter history, status-transition history, origin/reference info and vendor-change history are incorporated as the FL/server pieces are built - not retrofitted. No invoice-intake implementation now.

7. Reports reuse

New W M+ Reports consume the section-3 functions; they never reimplement the calculation.

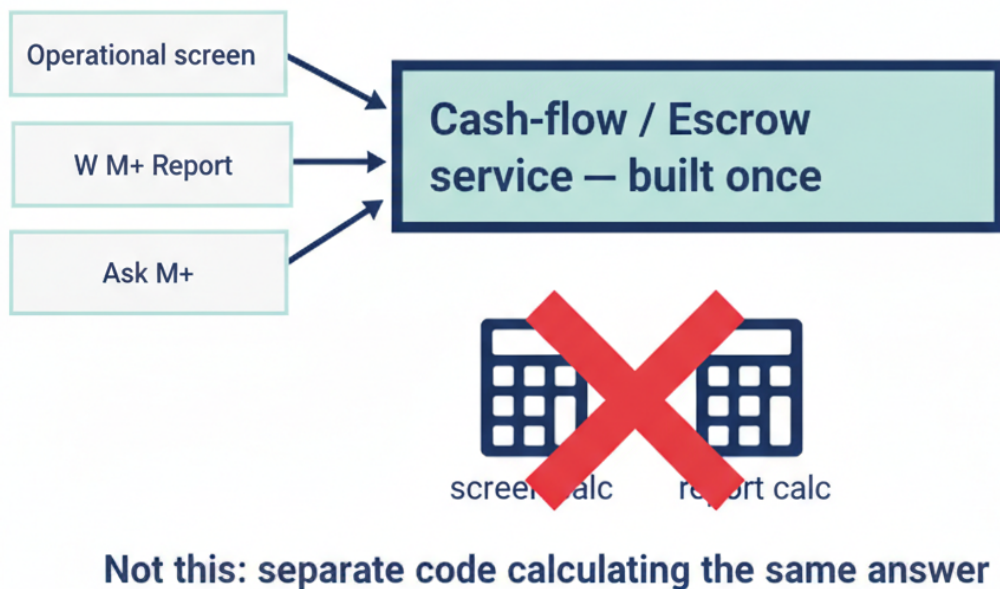


Fig. 5 - Reports consume the same service (Escrow model).

8. Track A work plan (proposed order)

- 1. Service framework: envelopes, errors, pagination/limits.
- 2. Tenant enforcement on (MgtCoClientID, HOALicenseNumber).
- 3. Date/period convention proposal -> sign-off.
- 4. Evidence/lineage in every envelope.
- 5. Event/status read access over existing data.
- 6. Seven Track A functions; getResidentAccountHistory first.
- 7. QA support: fixtures + per-function verification.

9. Out of scope

- FL-dependent functions and FL rule implementations.
- 12-level authorization catalog (awaiting spec).
- Invoice intake implementation.

10. Inputs needed

- Eight v1 questions: RECEIVED 2026-09-28, locked in section 3.
- Approval of this scope + section-5 conventions.
- Later: authorization catalog, then FL definitions per workflow.